

**Major Project Stage-1 Zeroth Review Presentation
on**

**Predictive Thermal Battery Management System
for Li-ion Battery Packs**

Presented by

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Abstract

- Lithium-ion cells are used in phones, laptops and electric vehicles, and they can catch fire if they are allowed to get too hot.
- Most small protection boards switch the pack off only after the temperature crosses a fixed value, usually about 60 °C.
- By the time a cell reaches that value it is often already heating itself, and that heat then causes still more heating.
- This project builds a battery management system for a four-cell lithium-ion pack that watches how fast the temperature is rising, not only how high it has reached.
- An ESP32 reads the cell voltages, the pack current and four cell temperatures, switches the pack off through MOSFETs when the rise is too fast, and sends every reading over Wi-Fi to a simple web dashboard.

Problem Statement

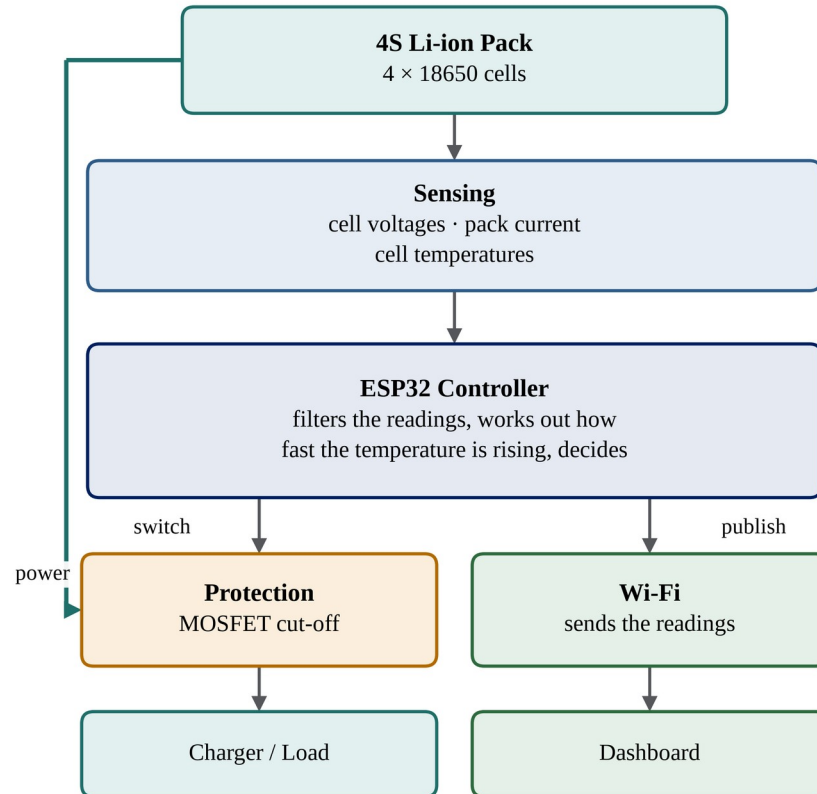
- A fixed temperature limit only acts once the cell is already hot, and it gives no warning before that point.
- A cell that is heating quickly at 35 °C is already faulty, but it passes a 60 °C test without any complaint.
- The sensor sits on the outside of the cell, so the inside is always hotter than the reading suggests.

Objective

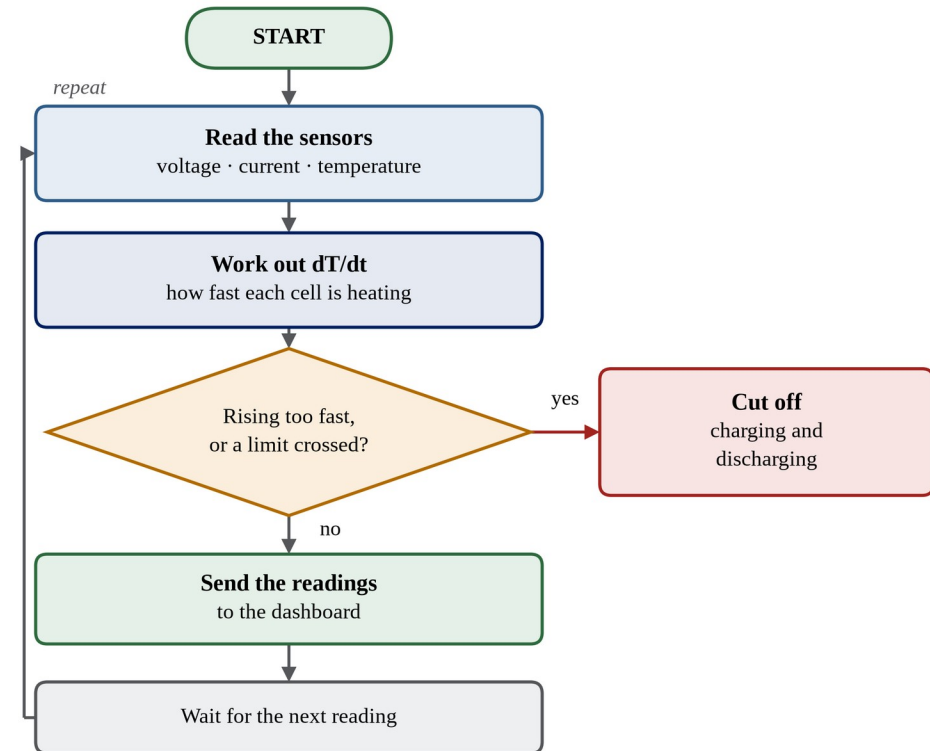
- To design and build a battery management system for a 4S lithium-ion pack that measures cell voltage, pack current and cell temperature, cuts off charging and discharging based on how fast the temperature is rising, and publishes every reading live over Wi-Fi.

Implementation Outline

Block diagram



Firmware flowchart



Implementation and Tools

This project has a hardware part and a software part. Both are built here.

Side	Tool	What it is used for
Hardware	Siemens NX	Modelling the enclosure
Hardware	3D printer, PLA+	Printing the enclosure
Hardware	Soldering station	Building the board, perfboard first, then a soldered PCB
Hardware	Multimeter, bench supply	Measuring and testing the circuit
Software	ESP-IDF	The framework the firmware is built on
Software	C++	The language the firmware is written in
Software	FreeRTOS	Running the sensor, safety and telemetry tasks
Software	HTML, CSS, JavaScript	The dashboard page in the browser
Software	Git and GitHub	Keeping the code and the documents

Literature Survey

References	Summary of the work
[1]	How thermal runaway starts inside a lithium-ion cell, and the temperature at which each stage begins.
[2]	A review of the ways thermal runaway can be predicted early, and what each method needs in order to work.
[3]	A warning method that acts when the temperature rise reaches 1 °C per second.
[4]	A review of what a battery management system does, and which problems in it are still open.
[5]	The Indian safety standard for sealed lithium cells, and the abuse tests it lays down.

References

- [1] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia and X. He, “Thermal runaway mechanism of lithium ion battery for electric vehicles: A review,” *Energy Storage Materials*, vol. 10, pp. 246–267, 2018.
- [2] X. Zhang, S. Chen, J. Zhu et al., “A critical review of thermal runaway prediction and early-warning methods for lithium-ion batteries,” *Energy Material Advances*, vol. 4, art. 0008, 2023.
- [3] Q. Chen, Y. He, N. Fang and G. Yu, “A combined data-driven and model-based algorithm for accurate battery thermal runaway warning,” *Sensors*, vol. 24, no. 15, art. 4964, 2024.
- [4] A. K. M. A. Habib, M. K. Hasan, G. F. Issa, D. Singh, S. Islam and T. M. Ghazal, “Lithium-ion battery management system for electric vehicles: Constraints, challenges and recommendations,” *Batteries*, vol. 9, no. 3, art. 152, 2023.
- [5] IS 16046 (Part 2) : 2018 / IEC 62133-2 : 2017, Secondary cells and batteries containing alkaline or other non-acid electrolytes — Part 2: Lithium systems.

Thank You

Questions are welcome